Spirit in Physics:

constructing the self as an information-geometric (tensor) space
from word association and physiology

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Abstract

We take one assumption seriously and build on it: the conscious self—"spirit"—*is* information. If information is physical (Shannon–Jaynes–Landauer), then a self made of information is a physical object, and the right way to study it is not to test a correlation but to **construct** it. We therefore give a **constructive** operational definition: the spirit of an individual *is* the information-geometric space we build from that person's own signals—word associations (Jung), reaction times, and physiological response—in which charged clusters (Jungian *complexes*) and their boundaries with concepts become geometric structure. The boundary of the self is, in principle, fixed physiologically by a rubber-hand-illusion paradigm; the conceptual structure is laid down by the word-association method. We render each individual as a third-order tensor (word $\times$ session $\times$ bodily-feature), embed it spectrally into a per-person "spirit space," and read complexes off as high-energy regions. **Information = physics = this mathematical space**: the model is not a description of the spirit, it is taken to be the spirit. In a pilot ($N = 5$ analyzed of 11; responses transcribed automatically) we build individual spirit spaces, locate their complexes, and use a tensor decomposition to look for a shared component—a candidate *collective unconscious*. It is faint but above chance (inter-subject $r = +0.07$, $p = 0.04$); its variance share is not estimable at $N = 5$ (the apparent $\sim$25% sits at the mechanical null). The spirit space is, at this scale, overwhelmingly individual. We present this as a reproducible construction and visualization (the full pipeline runs as a checkpointed StateGraph), not as a confirmatory study; the physiological boundary (RHI) data are the next installment.

1 Introduction

The concept of "spirit" has stayed outside empirical science for want of a physically grounded, measurable definition. Rather than ask whether spirit *can* be reduced to physics, we adopt a constructive stance: **assume** the self is information, build the object that assumption implies, and see what it looks like. If the assumption is wrong the construction will be empty or structureless; if it is right, an individual's spirit should appear as concrete geometry recoverable from that individual's own signals.

We keep the Japanese term *Jōcho* (情緒) untranslated for the layer of lived, meaning-laden experience not captured by "emotion" or "affect," following Kiyoshi Oka [12,13]. Our contribution is constructive and infrastructural, not a hypothesis test: (i) an operational identity—*the spirit of a person is the information-geometric space built from their associations and physiology*; (ii) a concrete construction of that space (tensor $\rightarrow$ spectral embedding) in which Jungian complexes are high-energy regions; and (iii) a decomposition that separates the individual space from a shared, collective component. The analytical-psychology vocabulary (Section 3) is an interpretive layer over the constructed geometry.

2 Theoretical Framework

We make three assumptions that turn "spirit" into a constructible physical object.

2.1 Assumption 1: Information is physical

Shannon (1948) defined information entropy H [11]; **Jaynes (1957)** showed it is formally the thermodynamic entropy S [4]; **Landauer (1961)** showed information processing has an energetic cost $k_B T \ln 2$ [6], experimentally verified [1]. We use this chain as the *motivating* grounding for treating an informational self as physical; we do not measure $k_B T$ or entropy production here.

2.2 Assumption 2: The self (spirit) is information

The **rubber-hand illusion** [2] shows the self-boundary is a malleable informational construct with a physiological signature—so the boundary of the self can, in principle, be *measured* (as a bodily response) rather than assumed. **Jung's word association** [5] shows the psyche is organized into *complexes*: clusters of charged concepts with measurable reaction times and autonomic responses. Together they license representing a self as charged structure in an informational space.

2.3 Assumption 3: Spirit as an open informational system

A self persists by exchanging information and entropy with its environment [9]; the **free-energy principle** [3] casts this as surprisal minimization. We adopt the resulting picture—a self is a structured region of an informational state space—and take the constructive step the principle invites: *build* that state space from data. Non-equilibrium thermodynamics [7,10] and feedback control [8] motivate, but are not invoked quantitatively in, the present pilot.

3 Structural Interpretation

We interpret the constructed geometry with analytical psychology:

- **Complex:** an energy-charged region of the individual's space (short or labile reaction time, elevated autonomic response).
- **Archetype:** a structural template recurring *across* individuals—hence a property of the shared component (Section 6).
- **Shadow:** suppressed, less reproducible parts of a complex.

4 Constructing the individual spirit space

We instantiate the spirit of a person as follows. Each stimulus word is presented (Jung's method); the spoken association is transcribed automatically (we reconstruct 851 stimulus→response pairs across the pilot from the recordings), and two bodily readouts are taken per trial: **response latency** (a temporal index of processing) and **skin potential** ΔSP (electrodermal response). The rubber-hand-illusion paradigm provides the bodily *boundary* of the self under congruent versus incongruent stimulation; those data are pending and the present construction uses the word-association and physiological channels.

For one individual, let each node w (a stimulus and/or its response concept) carry a bodily feature vector x_w (standardized latency and skin potential, per session). We define the **charge / energy** of a node intrinsically,

$$E(w) \;=\; z\big(\text{latency}(w)\big) + z\big(\Delta SP(w)\big), \tag{1}$$

build a similarity kernel between nodes, $W_{ij} = \exp(-\|x_i - x_j\|^2/\sigma^2)$ with σ the median pairwise distance, and embed the nodes by the low-lying eigenvectors of the symmetric normalized graph Laplacian $L_{\text{sym}} = I - D^{-1/2}WD^{-1/2}$. **This embedded, energy-weighted geometry is the spirit space of the individual.** We use two related notions of energy, and keep them distinct: the intrinsic *node charge* $E(w)$ of Eq. (1) (latency+arousal; this colours the charge geometry of Figs. 1–2), and the *manifold energy* of a node—its radius from the centroid of the spectral embedding—which is what ranks nodes into complexes; the two covary (within-individual $r \approx 0.52$ between node charge and manifold energy; both rise with latency and arousal). A *complex* is a high-energy region, and its boundary with neutral concepts is where the energy falls off. The construction is intrinsic—no cross-person normalization, no notion of how "common" a response is, enters it. We take this constructed object to be the spirit: information = physics = this space.

Well-posedness (a stipulative identity, not a tested hypothesis). To forestall a category error we state the logic explicitly. (i) "The spirit of a person *is* this space" is a **stipulative operational definition**, not an empirical claim to be confirmed by a p-value. (ii) What is empirically at stake is whether the construction is *non-empty*: a structureless result would show a flat Laplacian spectrum (no spectral gap), node energy uncorrelated with node identity, and no reproducible high-energy regions. (iii) We instead find structure on all three counts (Sections 5–6: spectral gap ≈ 0.19, participation ratio ≈ 10.9, low-rank Tucker geometry, reproducible complexes), so the construction is non-empty—the definition picks out something real, whose *scientific* content (individual structure, shared component) is then measured with appropriate restraint.

5 Pilot: individual spirit spaces and their complexes

We recruited 11 participants and analyze the $N = 5$ with synchronized, clock-aligned continuous skin potential. Across these, 970 trials were recorded; 945 have a valid skin-potential response window (used for the charge geometry) and 851 also carry a transcribed verbal response (used to label nodes). Each individual completed two test–retest sessions of an identical $\sim$100-word list. For each we build the spirit space of Section 4 and read off its energy geometry.

Complexes as high-energy regions. The energy geometry separates two intrinsic regimes (Fig. 1a): low-E attractor regions (fast, low-arousal, reproducible nodes—neutral concepts) and high-E regions (slow and/or high-arousal, less reproducible nodes), which are the candidate *complexes*. For the pilot cohort the high-energy region is populated by items such as *door* (ドア), *big* (大きい), *wash* (洗う), *habit* (癖), *expensive* (高価な), and the low-energy attractor by concrete neutral items—*ink* (インク), *tree* (木), *book* (本), *finger* (指), *dance* (踊る). These charged regions are exactly what Jung's complex predicts as measurable structure; here they are read directly from the body. Figure 2 shows the five individual spirit spaces—each a spectral geometry of that person's own nodes, coloured by energy—making concrete the claim that the constructed space *is* the individual's spirit. We quantify how individual they are by orthogonal-Procrustes disparity: between-person disparity (1.74) exceeds the within-person, session-to-session disparity (1.61) only marginally (ratio 1.08, bootstrap 95% CI [0.46, 1.08]—not reliably above 1). So although the panels look distinct, individual separation is *not statistically established* at $N = 5$; it is a target for the powered study, stated here without overclaim.

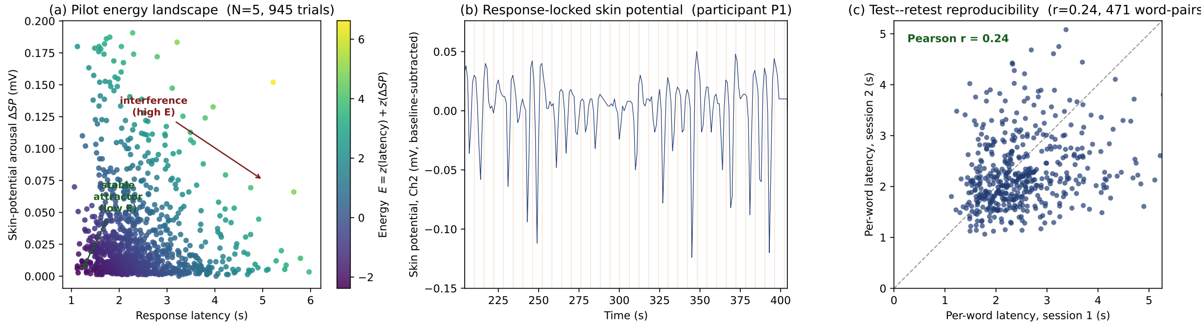

Figure 1: **The constructed spirit space (energy geometry).** (a) Every node is a word-association trial placed by response latency and skin-potential charge ΔSP and coloured by node energy E (Eq. (1)); low-E *attractor* regions (neutral concepts) and high-E *complex/interference* regions emerge intrinsically, with no popularity or frequency normalization. (b) Skin potential for one individual; vertical lines mark word onsets—the physiological signal from which the boundary is read. (c) Stability of the space across the two sessions (per-word latency, identical list; $r = 0.24$). Generated by `scripts/make_landscape_figure.py`.

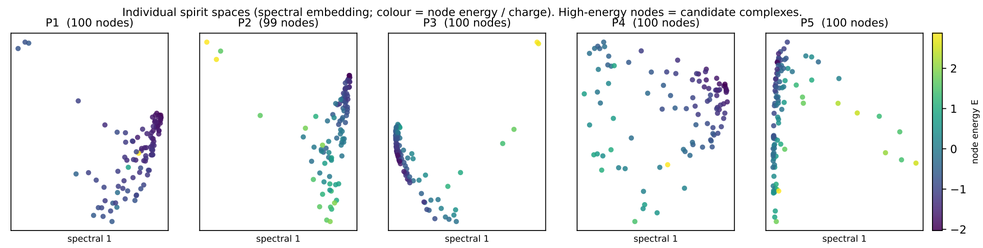

Figure 2: **Individual spirit spaces.** Each panel is one participant's own constructed space (spectral embedding of their word nodes from latency and skin potential), coloured by node energy E (Eq. (1)). The geometries are visibly distinct across individuals; high-energy (bright) nodes are that person's candidate complexes. Generated by `scripts/plot_individual_spirit.py`.

The space is dynamic and individual. Re-exposure (session $1 \to 2$) reshapes the space: response latency falls in 3/5 individuals (mean -0.25 s) and skin-potential charge in 4/5 (mean -0.004 mV), i.e. the geometry relaxes as concepts become familiar. The space is also strongly *individual*: per-word charge is only weakly shared across people (next section).

6 Individual vs. collective: tensor decomposition

To separate what is idiosyncratic to a person from what is shared, we stack the cohort as a third-order tensor $\mathcal{T} \in \mathbb{R}^{P \times W \times F}$ ($P = 5$ individuals, $W = 99$ shared words, $F = 3$ bodily features) and analyze its geometry; the full pipeline (build $\to$ tensorize $\to$ spectral $\to$ Tucker $\to$ summarize) runs as a checkpointed `langgraph-clj` StateGraph.

Shape of the shared space. A graph-Laplacian spectral embedding of the words (Fig. 4a,b) shows a **spectral gap** ≈ 0.19 and a participation ratio of ≈ 10.9 effective modes: the shared spirit space is genuinely multi-dimensional yet has a dominant collective coordinate. A Tucker (HOSVD) decomposition at ranks $[3, 6, 3]$ reconstructs 76% of the variance and saturates near 0.83 by word-mode rank ≈ 12 (Fig. 4c). Crucially the decomposition factorizes the tensor into a **participant mode (the individual spirit)** and **shared word-modes (structure common**

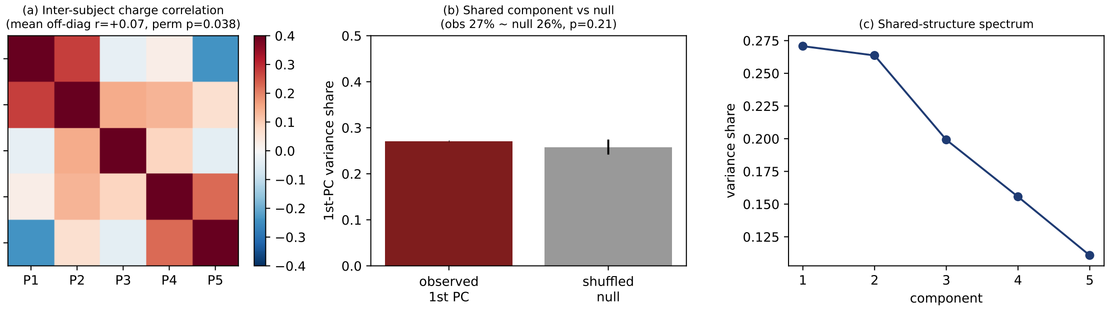

Figure 3: **Individual vs. collective structure.** (a) Inter-subject correlation of per-word charge (mean off-diagonal $r = +0.07$, permutation $p = 0.038$): a faint but above-chance shared signal on a largely private space. (b) The first-PC variance share (27%) sits at its shuffle-null baseline ($\approx$ 26%, $p = 0.21$)—i.e. a mechanical artifact of $P = 5$, *not* evidence of a large collective component. (c) The shared-structure spectrum. Generated by `scripts/plot_individual_spirit.py`.

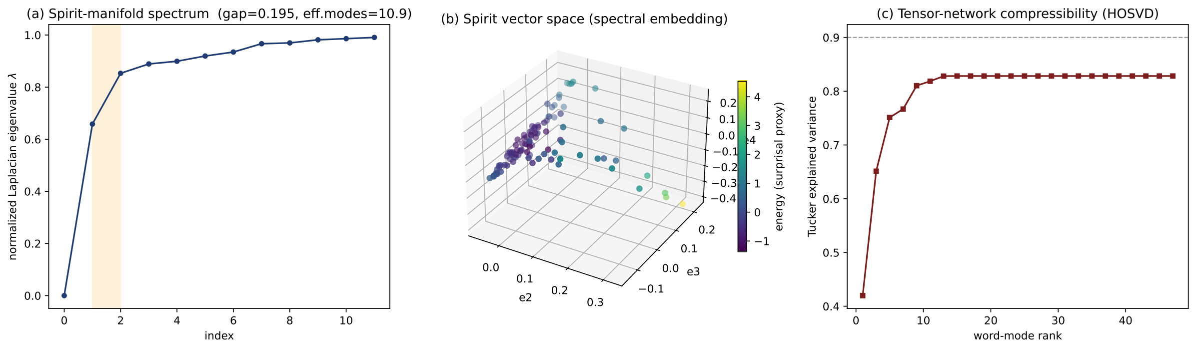

Figure 4: **The spirit space as geometry: individual and shared structure.** (a) Normalized graph-Laplacian spectrum; the gap (≈ 0.19) marks the dominant collective coordinate (effective modes ≈ 10.9). (b) Three-dimensional spectral embedding—the shared *spirit vector space*—with words coloured by energy. (c) Tucker (HOSVD) explained variance vs. word-mode rank; the tensor is low-rank (≈ 0.83 by rank ≈ 12) with an irreducible individual residual. These are descriptive geometry over $P = 5$, not inferential population statistics. Computed by the `langgraph-clj` StateGraph over `scripts/spirit_tensor.py`.

across people).

The collective component (a candidate collective unconscious). How much of the geometry is shared? We are deliberately conservative here, because with only $P = 5$ rows the first principal component mechanically absorbs a large share. Against a within-subject shuffle null: the inter-subject per-word charge correlation is weak but *above chance* ($\bar{r} = +0.065$, permutation $p = 0.038$), whereas the first-PC share (27%) is *indistinguishable* from its mechanical null ($\approx$ 26%, $p = 0.21$)—so the 27% is **not** evidence of shared structure, only the baseline. The honest reading is therefore narrow: there is a faint, above-chance shared component (the candidate *collective unconscious*), but its magnitude cannot be estimated at $N = 5$, and the variance-share figure should not be read as "25% collective." This shared component is the natural place to look for cultural/population structure (comparing tensor spaces across language or culture) in the powered study (Fig. 3).

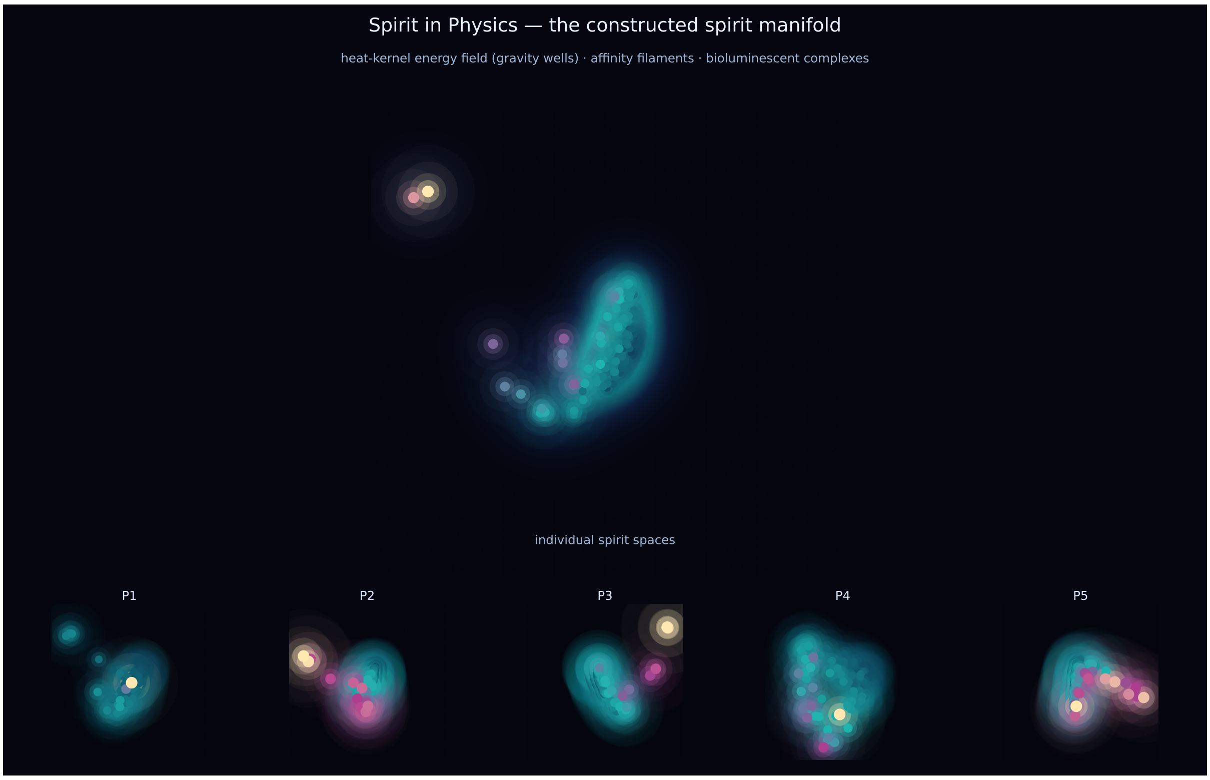

Figure 5: **The constructed spirit, rendered.** The shared spirit manifold (top) and the five individual spirit spaces (bottom), drawn from the real spectral embedding, node energy, and affinity kernel. The background is the heat-kernel energy field (curved-manifold "gravity wells"); threads are high-affinity edges; bright nodes are high-energy complexes. A visualization of the same object the paper constructs—no additional data. Generated by `scripts/render_spirit_awe.py`.

Visualization. Because the construction *is* the spirit, it is also something one can look at. Figure 5 renders the same quantities—spectral-embedding positions, node energy, and the affinity kernel—through the auxiliary physical pictures of Section 3: the energy landscape as a heat-kernel field (curved-manifold gravity wells), the kernel as luminous filaments, and complexes as bright bioluminescent concentrations, with each individual a distinct nebula beneath the shared manifold. The rendering is generated from the same pipeline (`scripts/render_spirit_awe.py`); it adds no data, only a faithful, high-altitude view of the constructed object.

7 Limitations

We state the scope plainly. (1) **Constructive, not confirmatory:** we build and visualize an object under the stated assumption; we do not claim to have tested "information = physics = spirit" as a hypothesis. (2) **Boundary data pending:** the rubber-hand-illusion measurement of the self-boundary—a core element of the method—is not yet collected; the present construction uses word association and physiology only. (3) **Power:** $N = 5$ (of 11). The shared/collective component is at most faint (inter-subject $r = 0.07$, $p = 0.04$; its variance share sits at the mechanical null), and per-person distinctness is not statistically established (Procrustes ratio 1.08, 95% CI $[0.46, 1.08]$). (4) **Descriptive geometry:** spectral gap, participation ratio and Tucker ranks are descriptive over $P = 5$, not inferential population statistics. (5) **Thermodynamics:** Landauer is a motivating analogy; no energetic quantity is measured. The powered study will add RHI boundary data, demographics for the cultural/collective analysis, and a pre-registered pipeline.

Data and code availability. All analysis code is bundled (`scripts/spirit_tensor.py`, `make_landscape_figure.py`, `plot_spirit_manifold.py`; the construction also runs as a `langgraph-clj` StateGraph) and regenerates every figure and number from the raw logs. For byte-stable results the analysis pins Python 3.14 / numpy 2.5 and fixed RNG seeds, so the reported permutation p-values (0.038, 0.21) and bootstrap CIs are deterministic; each script regenerates its outputs with a single invocation (e.g. `python scripts/spirit_tensor.py <stage> state.json`). Raw human-subjects recordings (face/voice, consent) contain personal information and are not publicly released; de-identified derived measures are available from the authors under a data-use agreement, consent permitting.

8 Conclusion

We treated "spirit" not as a hypothesis to confirm but as an object to construct: under the assumption that the self is information, an individual's spirit *is* the information-geometric space built from their associations and physiology, in which Jungian complexes appear as high-energy regions. A pilot ($N = 5$) builds such spaces, locates their complexes from the body alone, and finds the spirit space to be overwhelmingly individual with only a faint, above-chance shared component (candidate collective unconscious, $r = +0.07$, $p = 0.04$; its size not estimable at $N = 5$) via tensor decomposition. The construction is reproducible and the geometry is real; what remains for the powered, pre-registered study is the physiological boundary (rubber-hand illusion) and the cross-cultural reading of the shared space.

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